

Machine Learning: Code Overview & Oral Defense Guide

Forecasting European Housing Price Growth

1. Clustering Code Overview

Code section	What happens in the code?	Argumentation for this step
Setup and imports	The notebook loads the libraries for preprocessing, visualization, PCA, clustering, and validation.	This is the technical backbone of a structured pipeline. The key point is that clustering is part of a full workflow: prepare structural data, reduce dimensionality, compare clustering approaches, validate them, and then pass the labels into the forecasting stage.
Construction of country profiles	The original country-quarter panel is aggregated into one structural profile per country using means of key variables plus a volatility measure for house price growth.	This is appropriate because the paper clusters housing-market regimes, not quarterly observations. A country-level profile captures persistent structural characteristics such as growth, volatility, demographics, and financing conditions, which is exactly what is needed for regime-specific forecasting.
Initial profile inspection	The structural country matrix is inspected visually, for example with a heatmap.	This is useful because it checks whether meaningful cross-country heterogeneity is visible before formal clustering. In the paper, heterogeneity is a core assumption, so a descriptive inspection makes the later clustering step more credible.
Missing-value handling before PCA	Missing values in the structural profile matrix are filled so that PCA can be computed.	This is justified pragmatically because PCA requires a complete matrix. Since clustering is used to define broad structural regimes, a simple imputation step is acceptable as long as it is transparent and treated as preprocessing rather than as an economic claim.
Standardization	All structural variables are scaled before PCA and clustering.	Standardization is necessary because PCA and K-Means are distance- and variance-based. Without scaling, variables such as population density would dominate the geometry for purely numerical reasons. Scaling ensures that the cluster structure reflects cross-country patterns rather than measurement units.
PCA on country profiles	PCA is applied to the standardized country profiles to reduce correlated structural variables into a smaller number of components.	PCA is appropriate because many structural variables overlap conceptually and statistically. The goal is not to replace the original variables economically, but to build a cleaner and more compact space for clustering. That is consistent with the paper's aim of identifying broad structural axes rather than estimating causal effects at this stage.
Choice of number of PCA components	The notebook keeps four principal components based on explained variance.	Keeping four components is defensible because it preserves most of the structural information while avoiding an unnecessarily noisy clustering space. In the paper, four components explain 82.0% of the variance, so this is a reasonable balance between information retention and dimensionality reduction.
PCA loadings and biplot interpretation	The code inspects loadings and plots countries and variable directions in PCA space.	This step matters because PCA alone is too abstract. The biplot gives the components substantive meaning. In the paper, PC1 is interpreted as a dynamism axis and PC2 as a political-orientation/density axis, which makes the cluster structure economically interpretable and connects the code directly to the written argument.
Elbow and silhouette analysis	The notebook evaluates different K values for K-Means using inertia and silhouette scores.	This is the correct way to avoid choosing K arbitrarily. The elbow method shows where additional clusters stop improving compactness substantially, while silhouette assesses separation quality. In your data, silhouette remains weak and flat, so the correct conclusion is not that perfect natural clusters exist, but that the structure is gradual and K must be chosen as a practical partition.
Final choice of K = 4	K-Means is finally estimated with four clusters.	K = 4 is justified because it sits at the elbow and also matches the four-group housing taxonomy used in the literature-based comparison. That matters because the paper does not only partition countries statistically; it compares a data-driven four-cluster solution with a theory-grounded four-regime framework on equal footing.
Final K-Means clustering	The final K-Means model assigns each country to one of four clusters.	K-Means is appropriate here as a transparent benchmark for data-driven regime discovery. It is simple, interpretable, and easy to compare with the literature-based grouping. The paper does not claim that K-Means reveals true natural regimes, only that it provides a disciplined statistical alternative to theory-based classification.
Interpretation of K-Means cluster profiles	Cluster means are examined and translated into labels such as stagnation,	This step is justified because cluster labels are only useful if they become economically meaningful regimes. The names in the paper are not arbitrary; they are based on the profile table itself, for example low

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	stable core, convergence and growth, or structural outliers.	growth and high unemployment for stagnation, or high HPI growth and higher interest rates for convergence and growth.
DBSCAN parameter search and estimation	DBSCAN is applied on the PCA space across different epsilon values to test for dense natural groupings and noise points.	DBSCAN is included because it answers a different question than K-Means. K-Means always forces a partition, while DBSCAN tests whether the countries actually form dense natural groups. This is a strong design choice because the paper explicitly asks whether regimes are naturally data-driven or whether theory is needed to impose useful structure.
Interpretation of DBSCAN results	DBSCAN identifies one broad dense core and several countries that repeatedly appear as structural outliers.	This is one of the strongest findings in the code. It supports the paper's conclusion that European housing markets form more of a continuum than a set of sharply separated natural clusters. That explains why silhouette scores stay low and why theory-based stratification may outperform purely geometric clustering in forecasting.
Literature-based clustering	Countries are also assigned to theory-based groups such as West, North, South, and East.	This is justified because the research question is not which clustering algorithm is best, but whether structurally informed regimes improve forecasting. The literature-based grouping brings in institutional knowledge that PCA and K-Means cannot fully observe, such as housing systems, mortgage traditions, and welfare-state structures.
Comparison of K-Means and literature clusters	The code compares where both clustering approaches align and where they differ.	This comparison is analytically valuable because mismatches reveal the difference between statistical proximity and institutional similarity. If literature clusters forecast better despite weak natural separation, theory is adding useful structure beyond what the observed macro profiles alone can capture.
Internal validation metrics	Cluster quality measures such as silhouette and related diagnostics are calculated.	These metrics are useful, but they should be interpreted correctly. In this project they do not prove whether a regime is economically meaningful; they only describe the geometry of the cluster solution. Low silhouette scores are therefore not fatal and are instead read as evidence of a structural continuum.
External validation via the target variable	Cluster assignments are related back to next-quarter HPI growth to assess whether the clusters differ meaningfully in the forecasting target.	This is essential because the clustering only matters if it helps explain differences in house price dynamics. Linking regimes back to the target variable makes the unsupervised step forecasting-relevant. The logic is simple: if regimes differ in target dynamics, stratified models may learn more precise relationships.
ANOVA testing	The code tests whether target means differ across clusters.	ANOVA is appropriate here as a simple formal check that the clusters are associated with meaningful differences in average house price growth. It does not prove that the clustering is perfect, but it supports the claim that the regimes are not arbitrary.
Merge back into modelling data	Final cluster labels are merged back into the main dataset and exported for later forecasting.	This step is justified because the entire purpose of clustering in the paper is to enable pooled-versus-stratified forecasting. Merging the labels back into the modelling data operationalizes the clustering strategy and turns it into something testable in the later supervised phases.

In short: Most Important Clustering Questions and Short Answers

Important section	Short exam-ready answer
Country-profile construction	We constructed country profiles because we want to cluster housing market regimes , not quarterly observations. Aggregating to the country level captures persistent structural characteristics such as growth, volatility, demographics, and financing conditions.
Standardization before PCA and clustering	We standardized the variables because PCA and K-Means are scale-sensitive. Without standardization, high-scale variables like population density would dominate the geometry, and the clusters would reflect scale rather than structural similarity.
PCA and the choice of four components	We used PCA to reduce redundancy across correlated structural variables and create a cleaner clustering space. We kept four components because they retain most of the variance while keeping the reduced space compact and interpretable.
Choice of K = 4 for K-Means	We chose K = 4 because the elbow method supported it and because it aligns with the four-group literature taxonomy used elsewhere in the paper. So the choice is both empirically plausible and theoretically useful for comparison.
Use of DBSCAN in addition to K-Means	We added DBSCAN because K-Means always forces a partition, while DBSCAN tests whether natural dense clusters actually exist. This is important because our data looks more like a continuum with outliers than a set of sharply separated natural blocks.

Literature clusters as a parallel approach	We included literature-based clusters because the paper is not only about statistical grouping, but about whether theory-informed housing regimes improve forecasting. They bring in institutional structure that purely data-driven clustering may miss.
External validation and ANOVA	We validated the clusters against the target variable because clustering only matters if the regimes differ in house price dynamics. ANOVA supports that the groups are associated with meaningful differences in average target behavior, which makes them relevant for forecasting.

2. Modelling Code Overview

Code section	What happens in the code?	Argumentation for this step
Train-test split	The dataset is split temporally: pre-2022 observations are used for training, and 2022 onward is kept as the final test set.	This is the correct design because the task is genuine forecasting, not random prediction on shuffled data. A temporal split avoids leakage and makes the evaluation economically meaningful, especially since the post-2022 period contains the ECB rate-hiking shock that serves as a real structural-break test.
Walk-forward cross-validation	Within the training set, the code uses expanding-window TimeSeriesSplit for validation and hyperparameter tuning.	This is necessary because random k-fold CV would leak future information into the past. Walk-forward CV mirrors the actual forecasting problem: train on the past and validate on the next period. That makes the tuning process consistent with the paper's forecasting logic.
Benchmark model block	The code estimates three benchmark models: naive persistence, rolling mean, and OLS .	These benchmarks are essential because they set the reference level any ML model must beat. In the paper, this is especially important because the target has strong autocorrelation, so a naive benchmark is already competitive. Without these baselines, you could not judge whether model complexity adds real value.
Naive persistence model	The code predicts next-quarter HPI growth using the current quarter's growth.	This is justified because housing prices are persistent, and the EDA showed strong lag dependence. So naive persistence is not a dummy baseline, but a serious benchmark that captures autoregressive momentum. That is exactly why beating it is meaningful.
Rolling mean benchmark	The code predicts using the average of the previous four quarters.	This fits the data because the EDA showed persistence across several lags, including lag 4. The rolling mean therefore captures short-term cyclical dynamics while smoothing noise. It is a second simple but defensible time-series benchmark.
OLS benchmark	The code runs an ordinary least squares regression on the pooled feature set.	OLS is the simplest multivariate benchmark and shows how much can be explained by linear relationships alone before introducing regularization or non-linear ML. This matters because the later finding is that the best-performing pooled model is still basically linear.
Regularized linear model block	The code fits Ridge and Lasso models on the pooled data, using CV to tune the penalty.	This is justified because the feature space is relatively high-dimensional, includes multiple lags, and can contain correlated predictors. Regularization therefore improves stability and reduces overfitting, which is exactly why these models sit between OLS and tree-based methods in the modelling hierarchy.
Ridge regression	The code estimates Ridge with L2 regularization and tunes lambda via cross-validation.	Ridge is appropriate because it shrinks coefficients without dropping variables, which is useful when several macro predictors carry overlapping signal. In the paper, Ridge performs best among all pooled models, which supports the argument that the dominant relationships are mostly linear and benefit more from variance control than from extra non-linear flexibility.
Lasso regression	The code estimates Lasso with L1 regularization and tunes lambda via cross-validation.	Lasso is justified because it adds embedded feature selection. That makes it useful in a lag-rich macro panel where not every predictor is expected to matter equally. In the paper, Lasso removes several lagged variables, which supports the interpretation that only a subset of predictors contributes marginal forecasting value.
Tree-based model block	The code fits Random Forest and Gradient Boosting on the pooled data.	These models are included to test whether more flexible non-linear methods can outperform the linear benchmarks. This is theoretically motivated by macro forecasting literature, but the results show that extra non-linear capacity is not enough to overcome regime instability in the pooled setting.
Random Forest	The code estimates a Random Forest with tuned settings such as number of trees and leaf size.	Random Forest is appropriate because it can capture non-linearities and interactions without strong functional-form assumptions. It is a sensible test of whether housing price dynamics contain richer structure than linear models can capture. In the paper, however, it does not beat Ridge in the pooled setting, which supports the conclusion that non-linearity is not the main bottleneck.
Gradient Boosting	The code estimates Gradient Boosting with tuned hyperparameters such as learning rate, depth, number of estimators, and subsampling.	Gradient Boosting is justified because it is one of the strongest standard ML methods for structured tabular data and can learn incremental corrections to residual errors. This makes it especially relevant in a forecasting setup where subtle regime-specific patterns may exist. In the results, it becomes most valuable in the stratified setting, especially for the North cluster.
Pooled modelling phase	The code trains all benchmark and ML models once on the full pooled dataset.	This pooled phase is necessary because it gives the baseline against which all stratified models are judged. Without it, you could not answer

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		whether regime-specific modelling adds value. In the paper, pooled models improve RMSE only modestly and all have negative R^2 , which is interpreted as evidence of regime instability rather than pure model failure.
Cluster-stratified modelling phase	The code repeats the model estimation separately within each K-Means cluster and each literature-based cluster.	This is the core methodological test for RQ2. The logic is that a model trained only within a more homogeneous housing regime can learn a cleaner relationship between inputs and target than a pooled model that averages across very different institutional systems.
Data-driven stratified models	The code fits the selected models separately inside the K-Means clusters.	This is justified as the pure data-driven version of regime-specific forecasting. It tests whether the macro-financial structure alone is sufficient to define useful forecasting groups. In the paper, the answer is mixed, which is itself informative: statistical proximity in PCA space is not always enough to produce coherent predictive regimes.
Literature-based stratified models	The code fits the selected models separately inside the West/North/South/East literature clusters.	This is justified because the paper explicitly hypothesizes that institutional knowledge may outperform purely data-driven grouping when natural cluster separation is weak. The results support that logic: literature-based stratification performs better in most key cases, especially North, West, and East.
Prediction collection and comparison	The code stores all out-of-sample predictions and compares them in one common results table.	This is important because the paper is comparative by design. The value is not just estimating models, but evaluating them under the same target, same split, and same metrics. That makes the final conclusions about pooled versus stratified modelling defensible.
Evaluation metrics	The code computes RMSE, MAE, and R^2 for all models and model-cluster combinations.	These metrics are chosen for complementary reasons. RMSE is the main metric because large forecast errors matter economically in housing markets. MAE gives a more robust view less driven by extremes. R^2 helps diagnose whether the model adds value over a simple mean benchmark.
RMSE as primary metric	The code ranks models primarily by RMSE.	RMSE is the right primary metric here because housing booms and downturns make large errors especially costly and analytically important. The paper therefore treats RMSE as the main performance criterion rather than R^2 alone.
Interpretation of negative R^2	The code reports R^2 even when it is negative on the test set.	Keeping negative R^2 is important because it is diagnostically honest. In the paper, negative R^2 is not hidden; it is interpreted as evidence that the post-2022 regime shift broke the stability of pooled relationships.
Model ranking and pooled results table	The code produces the final pooled comparison table showing naive, OLS, rolling mean, Ridge, Lasso, Random Forest, and Gradient Boosting.	This is justified because the paper's first research question is explicitly comparative. The pooled ranking supports the substantive conclusion that Ridge performs best and that added complexity does not automatically improve forecasting in this panel.
Cluster-level performance comparison	The code compares pooled versus stratified performance within each cluster.	This is the decisive empirical test of whether stratification helps. It is justified because the paper's contribution is not simply that it clustered countries, but that it tested whether clustering improves forecasting. The results show that the answer depends on regime coherence and sample size.
Feature importance block	The code extracts feature importance, especially from Random Forest models, for pooled and regime-specific analysis.	This is justified because the paper's third research question asks which variables drive short-term housing price dynamics. Tree-based importance is not causal, but it is a useful model-based summary of predictive relevance. In the results, interest rates and HPI lags dominate, which aligns with the theory and the EDA.
Pooled feature importance interpretation	The code shows which predictors matter most in the pooled Random Forest.	This matters because it translates the model output back into an economic story. In the paper, the dominant role of lagged long-term interest rates and autoregressive HPI lags supports the view that housing markets are driven by monetary conditions and momentum effects.
Regime-specific feature importance	The code compares feature importance across literature-based housing regimes.	This is justified because one of the paper's claims is that housing price dynamics are regime-dependent rather than universal. Showing different importance rankings across West, North, South, and East provides exactly that evidence.
Runtime / complexity comparison	The code records model training times and compares performance against computational cost.	This is important because the paper does not only ask which model predicts best, but also whether the gain is worth the complexity. The results support the conclusion that simple regularized linear models offer the best efficiency-performance trade-off in this application.
Final interpretation layer	The code outputs are used to support the paper's final conclusions on modest pooled ML gains, stronger literature-	This is justified because the modelling notebook is not an isolated exercise. It directly produces the evidence for RQ1, RQ2, and RQ3. The code is structured to answer the research questions step by step, not just to maximize predictive performance.

Code section	What happens in the code?	Argumentation for this step
	based stratification, and the central role of interest rates.	

In short: Most Important Modelling Questions and Short Answers

Important section	Short exam-ready answer
Temporal train-test split	We used a temporal split because this is a genuine forecasting task. Training on pre-2022 data and testing on 2022 onward prevents data leakage and makes the evaluation realistic under the post-2022 structural break.
Walk-forward cross-validation	We used walk-forward cross-validation because hyperparameter tuning must preserve time order. It mimics the real forecasting problem by always training on the past and validating on the next period.
Strong benchmark models	We included strong benchmarks because the target is persistent, so ML only matters if it beats serious baselines. Naive persistence and rolling mean are therefore necessary reference points, not trivial comparisons.
Ridge and Lasso as core pooled models	We used Ridge and Lasso because the pooled panel contains many lagged and partly correlated predictors. Regularization controls variance and overfitting, which is why these models are more appropriate than plain OLS in the pooled setting.
Pooled versus stratified modelling	We compare pooled and stratified models because this is the central empirical test of the paper. The whole question is whether housing-market regimes allow models to learn more precise relationships than one common pooled specification.
Literature-based versus data-driven stratification	We compare both because the paper asks whether institutional knowledge adds value beyond statistical grouping. The results show that theory-informed clusters are more useful when natural separation in the data is weak.
Negative R^2 in the pooled test period	We interpret negative R^2 as evidence of regime instability rather than simple model failure. The post-2022 rate-hiking cycle changed housing-market dynamics so strongly that pooled models trained on the earlier low-rate period did not generalize well.
Feature importance by pooled and regime-specific models	We analyze feature importance because it links prediction back to the economic question of what drives short-term house price growth. In our results, interest rates and HPI lags dominate, and their importance varies across regimes.
Runtime versus performance trade-off	We compare runtime and performance because the best model is not only the most accurate one, but the one that delivers useful gains relative to its added complexity. In this paper, simple regularized linear models are more efficient than more expensive tree-based models.